Supermarket Sales Analysis

Excel dashboard project | Jan to Mar 2019, 3 branches

Goal

Find where and when a three-branch supermarket makes the most money, and what keeps customers happy.

Data

1,000 sales from three branches (A, B and C) between January and March 2019. Each row is one sale, with product line, quantity, unit price, tax, payment method, customer type, time of day and a customer rating from 4 to 10.

What I did

- **Cleaned the data.** Checked for blank cells and duplicate invoices (none found), confirmed price × quantity + 5% tax matched on every row, and fixed a hidden trailing space in a column header.
- **Relabelled values.** Turned codes into readable labels (Member / Non-member, Male / Female).
- **Added columns.** Month, Time Slot (Morning, Afternoon, Evening, Night) and Rating Group.
- **Built the dashboard.** Pivot tables, six charts and five KPI cards, all linked to Branch and Month slicers.

Key numbers

Measure	Value
Total sales	322,966.75
Number of sales	1,000
Average sale	322.97
Average rating	6.97
Total units sold	5,510

Findings

- **Afternoons drive the business.** Afternoon sales total 148,023 out of 322,967, about 46% of revenue.
- **Branches are very close.** A: 106,200 | B: 106,198 | C: 110,569. Branch C leads by only about 4%.
- **Members spend slightly more.** Members account for 51% of sales (164,223) and non-members 49% (158,743).
- **January was the strongest month.** January 116,292 | February 97,219 | March 109,456. February is a shorter month, so the gap should be read with care.
- **Product lines are fairly even.** Food and beverages leads at 56,145, and Health and beauty is lowest at 49,194.
- **Payment methods are close.** Cash 344 sales, e-wallet 345, credit card 311.

Limits of the data

- There are no customer IDs, so I counted sales, not customers.
- There is no real cost data. The "gross income" column equals the tax amount and "gross margin percentage" is a constant 4.76%, so profit cannot be measured.
- Only three months of data, so seasonal patterns cannot be judged.

Recommendations

- Schedule more staff and run promotions in the afternoon, when most revenue is earned.
- Test a member-only offer to see whether it lifts spending.
- Collect customer IDs and cost data so future analysis can cover repeat customers and profit.

Tools

Microsoft Excel: Tables, PivotTables, PivotCharts, slicers, formulas (IF, HOUR, SUM, COUNTA, AVERAGE, GETPIVOTDATA).